

DS 498-02

R Practice 2

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INSTRUCTIONS: R Practices have the purpose of making students comfortable with the basics of R and R Studio for empirical economic analysis.

In this practice, you will improve your **intuition** about the main {tidyverse} functions: `select()`, `filter()`, `mutate()`, and `group_by()`. Not only is it important to know how to apply them in your own tasks, but also to **interpret** what a *pipeline* is doing when you are manipulating your data.

Have fun!

Problem 1

Suppose you are working with an R object called `price_data` in your RStudio environment. (This file does not exist in reality.) It looks like the following:

```
price_data
```

```
## # A tibble: 9 × 3
##   year price state
##   <dbl> <dbl> <chr>
## 1  2006   2.5  NY
## 2  2007    3   LA
## 3  2008   3.15 CA
## 4  2015    2   ID
## 5  2006   1.15 VT
## 6  2008   2.75 NY
## 7  2008    4   CT
## 8  2007   5.15 CT
## 9  2015    2   NY
```

For the following 5 parts, write down, in plain English, what each pipeline is trying to extract from this data set:

(a)

```
price_data %>%
  select(year, price)
```

(b)

```
price_data %>%
  filter(year > 2008)
```

(c)

```
price_data_subset ← price_data %>%
  filter(state %in% c("NY", "CT"))
```

(d)

```
price_data %>%
  mutate(price_cents = price * 100)
```

(e)

```
price_data %>%
  group_by(year) %>%
  summarize(mean_price = mean(price))
```